Forces on two objects

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify an equal-and-opposite interaction pair on different objects.

SUCCESS CRITERIA - I can...

1. Name both objects in an interaction pair.
2. Place the two equal-and-opposite forces on different objects.
3. Explain why equal forces can produce unequal accelerations.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “interaction” and “system” to explain your answer. Sentence starters are optional.

interaction: two objects affecting one another

system: the object or objects chosen for analysis

acceleration: rate of change of velocity

01 Warm-up: bring back an earlier idea

With a fixed net force, a **4 kg** system accelerates at **12 m/s²**.
1. **Predict acceleration** for a **8 kg** system.
2. **State a control needed** for this comparison.

Sentence starter: I predict … . I used … to make this prediction because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $F_{A \text{ on } B} = -F_{B \text{ on } A}$.
The two forces act on different objects at the same instant. Their accelerations can differ.

Worked reasoning: Two skaters push apart. Each feels **20 N** from the other. If their masses are **40** and **80 kg**, accelerations are **0.5** and **0.25 m/s²** in opposite directions.

Your turn: 1. **Draw separate force diagrams** for the horizontal interaction.
2. **Explain why equal forces** do not imply equal accelerations.

Sentence starter: I think … . I can point to …, which shows …, so … .



Skater A: 40 kg



Skater B: 80 kg

Marzano check - circle: ① ② ③

03 Name both forces

A person presses a wall to the right. On the two-object diagram, draw the force the hand exerts on the wall and the force the wall exerts on the hand. **Label both forces** with their source and receiving object.

Sentence starter: In my work, I used … to show … . I labeled it … .



Hand



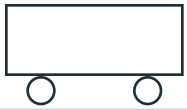
Wall

Marzano check - circle: ① ② ③

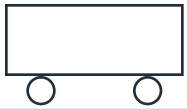
04 Heavy and light carts

A heavy cart and a light cart push apart on a low-friction track.
1. **Add their interaction-force arrows** to the diagram.
2. **Compare force sizes** and **explain which cart accelerates more**.

Sentence starter: Both … . They differ in … because … .



Light cart



Heavy cart

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Balanced or a pair?

A book rests on a table. Earth pulls book down; table pushes book up. Are these the two forces of one interaction pair? Explain by naming the object receiving each.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Start at **6 m**, move to **16 m**, then finish at **8 m**. **Draw the route**. **Find distance**, displacement with direction and final position.
Label the diagram with the given positions and any prediction.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I predict … . I used … to make this prediction because … .

Route



Position (m)

Marzano check - circle: 1 2 3

07 Exit check

A runner's foot pushes backward on the ground without slipping.
1. **Add the two interaction-force arrows** to the diagram.
2. **Name the partner force** and the object it acts on.
3. **Explain how the runner can** accelerate forward.

Sentence starter: I think … . I can point to …, which shows …, so … .



Foot



Ground

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. **Choose two masses** that would give a **3**: **1** acceleration ratio while the forces remain equal.
2. **Explain which skater accelerates more**.

Sentence starter: I think … . I can point to …, which shows …, so … .

Skater	Mass (kg)	Acceleration ratio
A	Choose: ____	3
B	Choose: ____	1

Marzano check - circle: 1 2 3